

Structure from Motion

Lecture 15

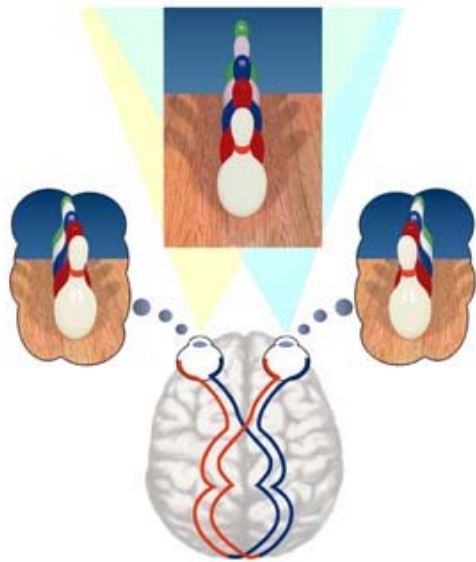
Shape From X

- Recovery of 3D (shape) from one or two (2D images).

Shape From X

- Stereo
- Motion
- Shading
- Photometric Stereo
- Texture
- Contours
- Silhouettes
- Defocus

Shape from Stereo



(a)



(b)

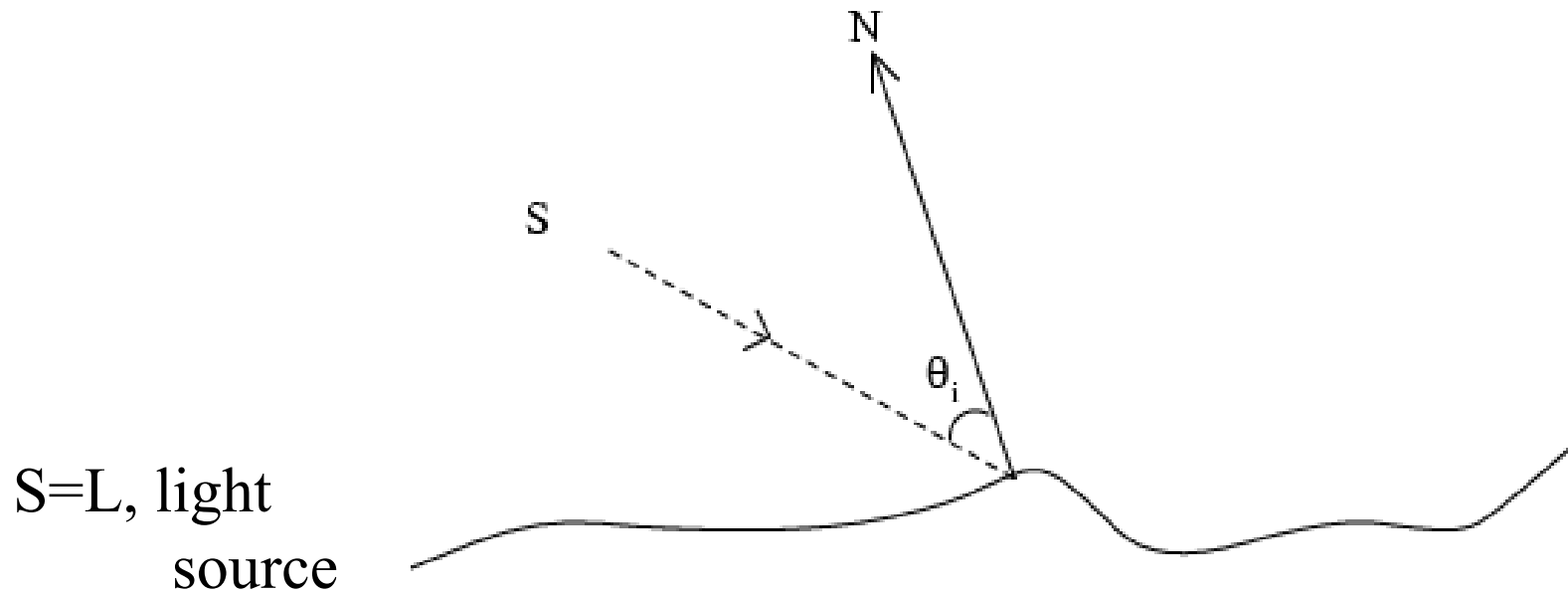


(c)

Shape from Shading



Lambertian Model



$$f(x, y) = n.L = (n_x, n_y, n_z).(l_x, l_y, l_z)$$

$$f(x, y) = n.L = \left(\frac{1}{\sqrt{p^2 + q^2 + 1}} (-p, -q, 1) \right).(l_x, l_y, l_z)$$

Sphere

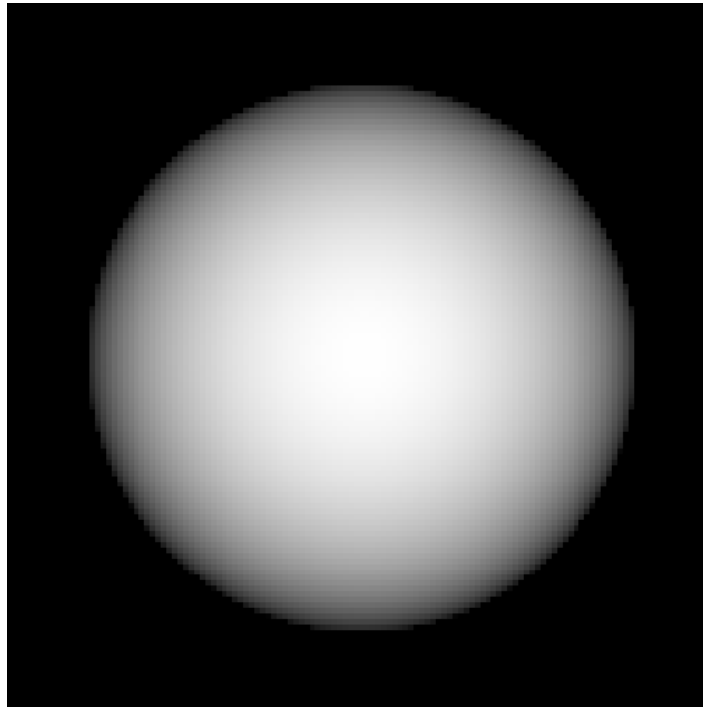
$$z = \sqrt{(R^2 - x^2 - y^2)}$$

$$p = \frac{\partial z}{\partial x} = -\frac{x}{z}$$

$$q = \frac{\partial z}{\partial y} = -\frac{y}{z}$$

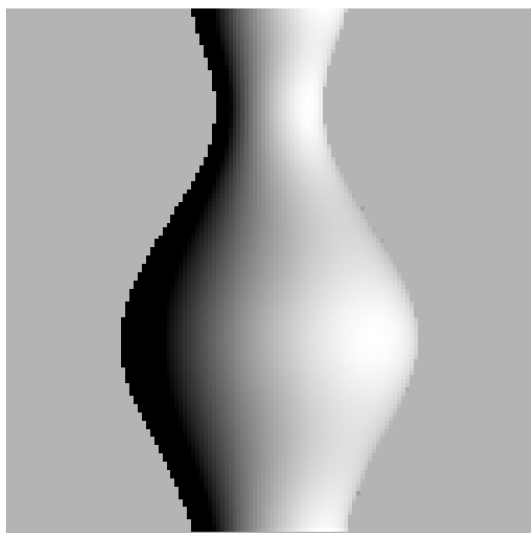
$$(n_x, n_y, n_z) = \frac{1}{R}(x, y, z)$$

Sphere

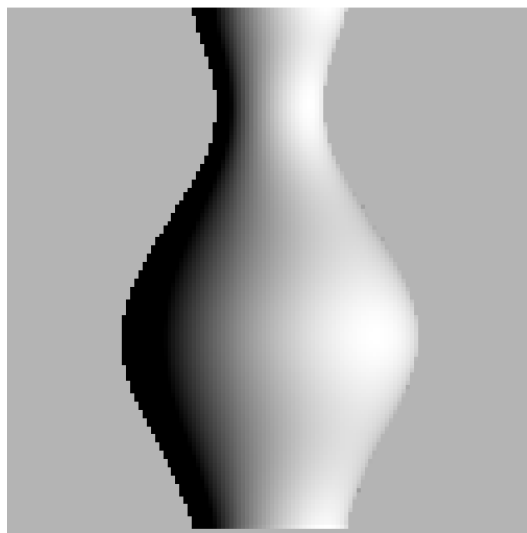


Copyright Mubarak Shah 2003

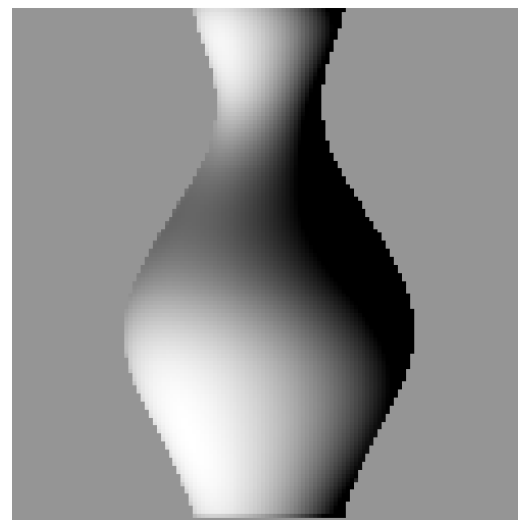
Vase



$(1, 0, 1)$



$(-1, 1, 1)$



$(-1, -1, 1)$

Shape from Shading

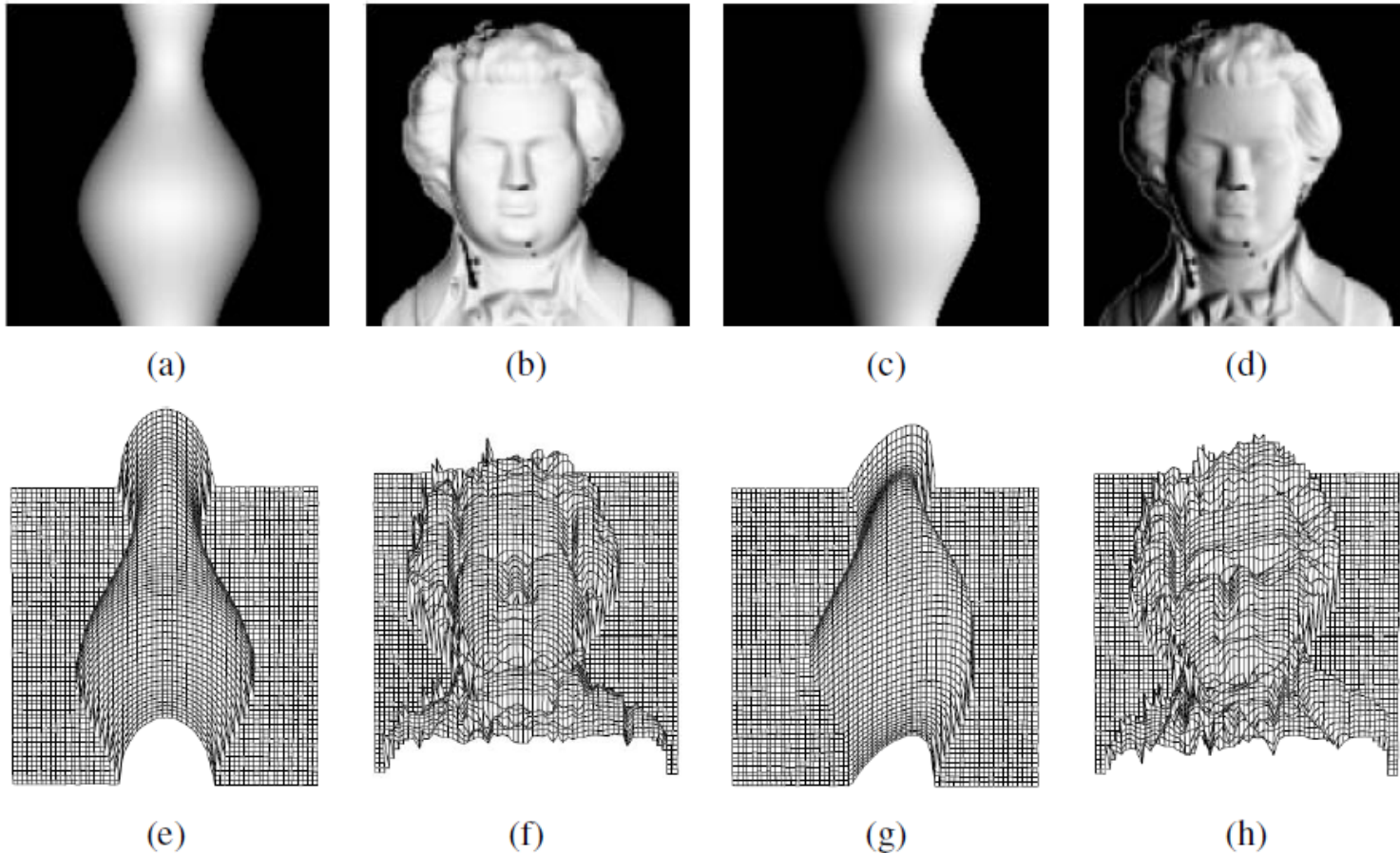


Figure 12.2 Synthetic shape from shading (Zhang, Tsai, Cryer *et al.* 1999) © 1999 IEEE: shaded images, (a–b) with light from in front $(0, 0, 1)$ and (c–d) with light the front right $(1, 0, 1)$; (e–f) corresponding shape from shading reconstructions using the technique of Tsai and Shah (1994).

Photometric Stereo

- Use images taken under different illumination conditions to recover 3D shape.

Photometric Stereo

$$I(x, y) = B \vec{N} \cdot \vec{S},$$

B= Albedo

N=surface normal

S=light source direction

Take Three images under different light sources

$$I_1(x, y) = B \vec{S}^1 \cdot \vec{N},$$

$$I_2(x, y) = B \vec{S}^2 \cdot \vec{N},$$

$$I_3(x, y) = B \vec{S}^3 \cdot \vec{N}.$$

$$B \begin{bmatrix} s_x^1 & s_y^1 & s_z^1 \\ s_x^2 & s_y^2 & s_z^2 \\ s_x^3 & s_y^3 & s_z^3 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}.$$

Photometric Stereo

$$B \begin{bmatrix} s_x^1 & s_y^1 & s_z^1 \\ s_x^2 & s_y^2 & s_z^2 \\ s_x^3 & s_y^3 & s_z^3 \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} = \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix}.$$

$$B A \vec{N} = I$$

$$B \vec{N} = A^{-1} I,$$

$$B ||\vec{N}|| = ||A^{-1} I||$$

$$||n|| = 1 \quad B = ||A^{-1} I||.$$

$$\vec{N} = \frac{A^{-1} I}{B}$$

Results

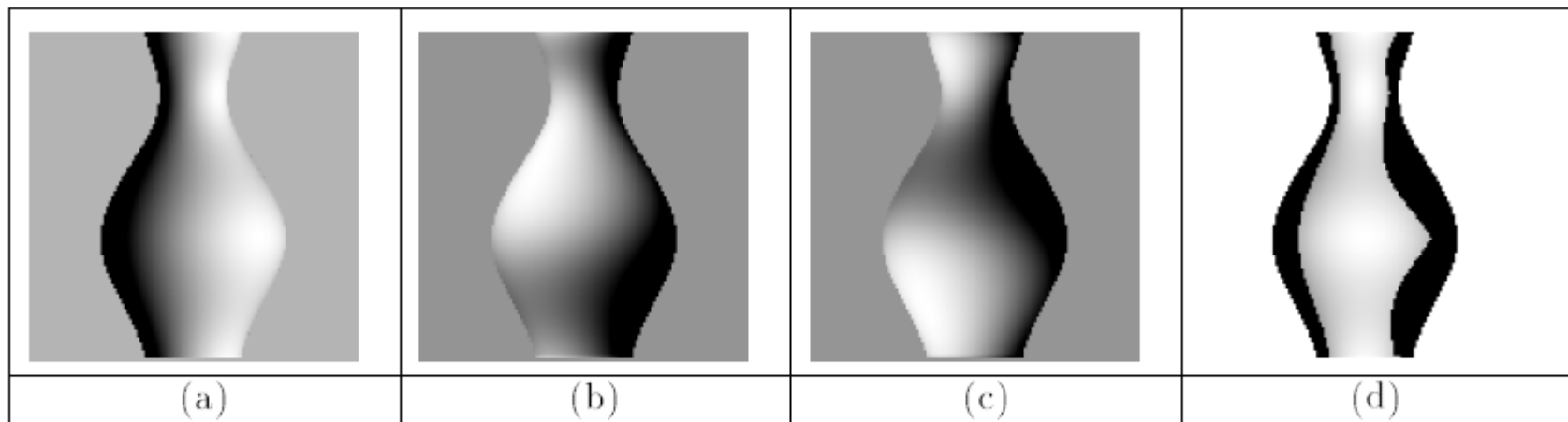
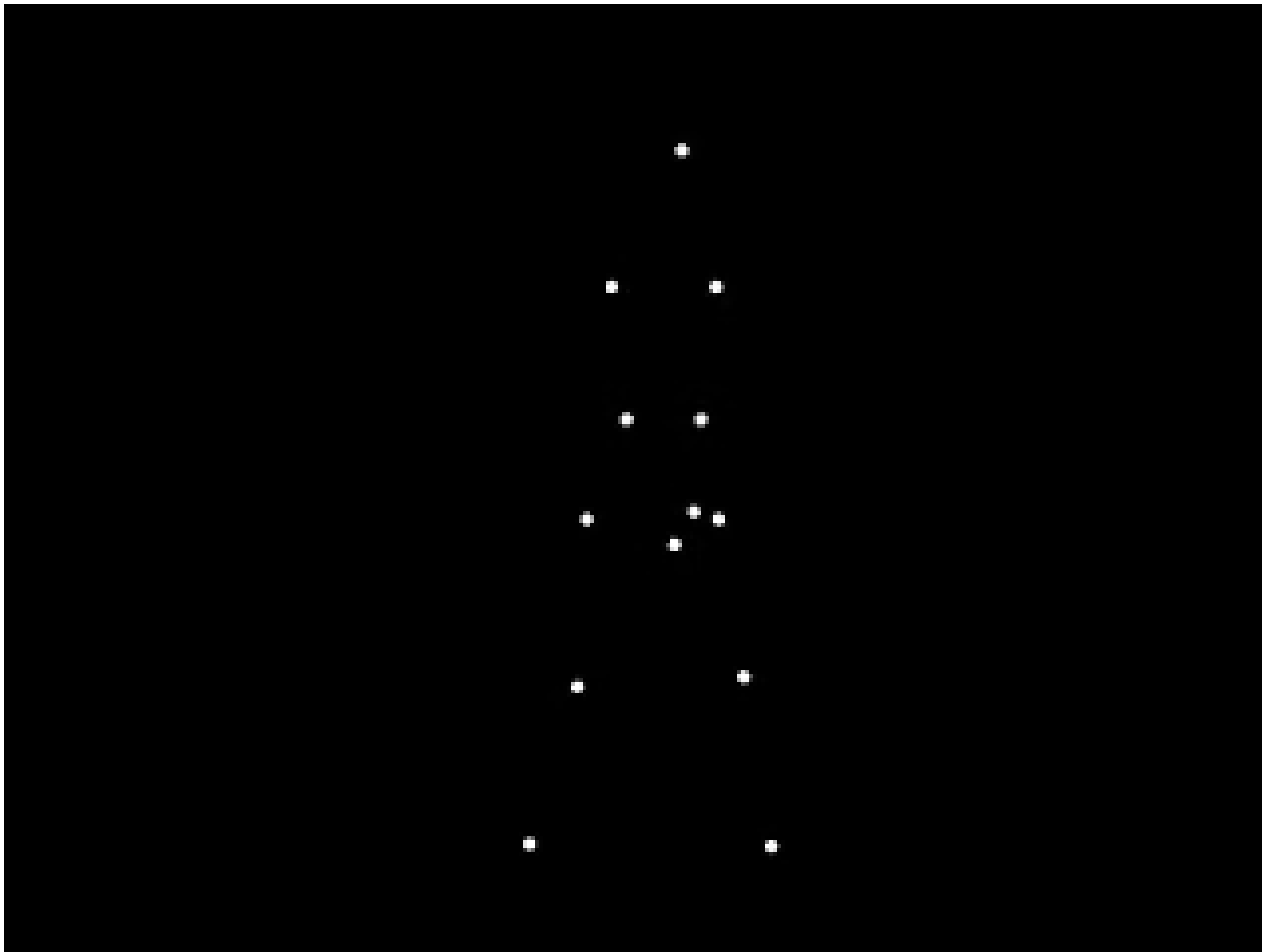


Figure 6.13: (a)-(c) Input images with light sources $(1, 0, 1)$, $(-1, 1, 1)$, $(-1, -1, 1)$ (d) Intensity image reconstructed using the surface normals computed by photometric stereo.

Structure from Motion

Copyright Mubarak Shah 2003

Shape from Motion: Moving Light Display



Shape from Motion



(a)



(b)



(c)



(d)

Problem

- Given optical flow or point correspondences, compute 3-D motion (translation and rotation) and shape (depth).

Structure from Motion

- S. Ullman
- Hanson & Riseman
- Webb & Aggarwal
- T. Huang
- Heeger and Jepson
- Chellappa
- Faugeras
- Zisserman
- Kanade
- Pentland
- Van Gool
- Pollefeys
- Seitz & Szeliski
- Shahsua
- Irani
- Vidal & Yi Ma
- Medioni
- Fleet
- Tian & Shah

Tomasi and Kanade Factorization Orthographic Projection

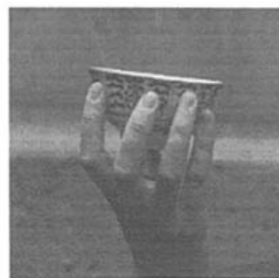
Assumptions

- The camera model is orthographic.
- The positions of “p” points in “F” frames ($F \geq 3$), which are not all coplanar, and have been tracked.
- The entire sequence has been acquired before starting (batch mode).
- Camera calibration not needed, if we accept 3D points up to a scale factor.

Input



1



100



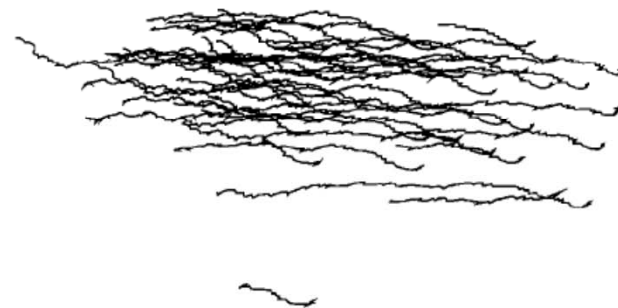
150



210

(a)

Images



(b)

KLT Tracks

Feature Points

Image points $\{(u_{fp}, v_{fp}) \mid f = 1, \dots, F, p = 1, \dots, P\}$

$$W = \begin{bmatrix} u_{11} \dots u_{1P} \\ \vdots \\ u_{F1} \dots u_{FP} \\ v_{11} \dots v_{1P} \\ \vdots \\ v_{F1} \dots v_{FP} \end{bmatrix} \qquad W = \begin{bmatrix} U \\ - \\ V \end{bmatrix}$$

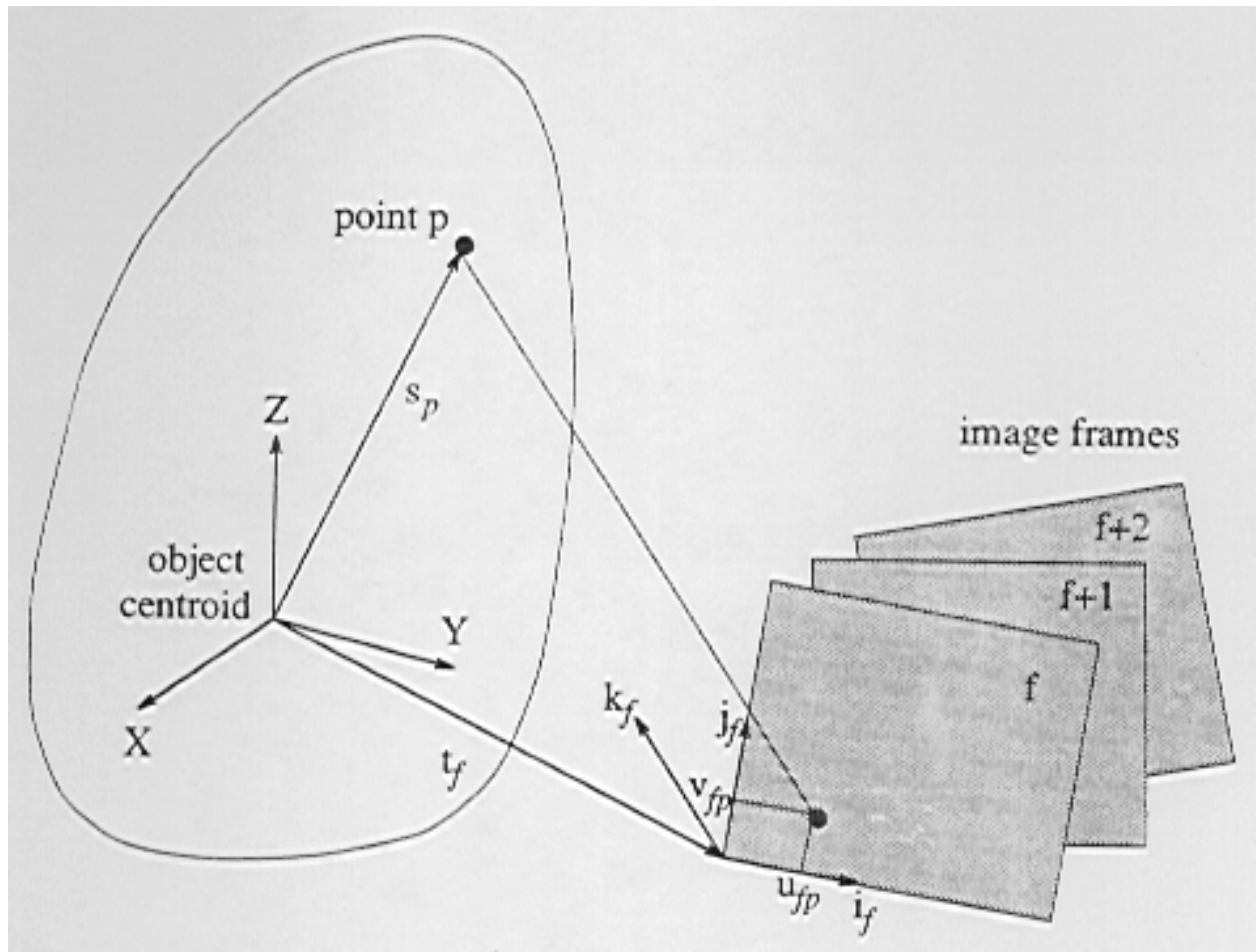
Mean Normalize Feature Points

$$a_f = \frac{1}{P} \sum_{p=1}^P u_p \qquad b_f = \frac{1}{P} \sum_{p=1}^P v_p$$

$$\tilde{u}_{fP} = u_{fP} - a_{fP} \quad (\text{A})$$

$$\tilde{v}_{fP} = v_{fP} - b_{fP}$$

Orthographic Projection



Copyright Mubarak Shah 2003

Orthographic Projection: Feature Points

$$s_p = (X_p, Y_p, Z_p)$$

3D world
point

$$u_{fP} = i_f^T (s_P - t_f) \quad (\text{C})$$

Orthographic
projection

$$v_{fP} = j_f^T (s_P - t_f)$$

$$k_f = i_f \times j_f \quad i, j, k \text{ are unit vectors along } X, Y, Z$$

Orthographic Projection: Mean Normalized Feature Points

$$\tilde{u}_{fp} = u_{fp} - a_f \quad a_f = \frac{1}{P} \sum_{n=1}^P u_p$$

$$= i_f^T (s_p - t_f) -$$

$$= i_f^T \left[s_p - \frac{1}{P} \sum_{q=1}^P s_q \right]$$

$$= i_f^T s_p$$

If Origin of world is at the centroid of object points

Orthographic Projection: Mean Normalized Feature Points

$$\begin{aligned}\tilde{u}_{fp} &= u_{fp} - a_f & a_f &= \frac{1}{P} \sum_{p=1}^P u_p \\ &= i_f^T (s_p - t_f) - \frac{1}{P} \sum_{q=1}^P i_f^T (s_q - t_f) \\ &= i_f^T \left[s_p - \frac{1}{P} \sum_{q=1}^P s_q \right]\end{aligned}$$

If Origin of world is at the centroid of object points

Registered Measurement Matrix

$$\tilde{\mathbf{u}}_{fP} = \dot{\mathbf{i}}_f^T \mathbf{S}_P$$

$$\tilde{\mathbf{v}}_{fP} = \dot{\mathbf{j}}_f^T \mathbf{S}_P$$

$$\tilde{\mathbf{W}} = \begin{bmatrix} \tilde{\mathbf{U}} \\ - \\ \tilde{\mathbf{V}} \end{bmatrix}$$

Registered Measurement Matrix

$$\tilde{u}_{fP} = \dot{i}_f^T S_P \quad (\text{B})$$

$$\tilde{v}_{fP} = \dot{j}_f^T S_P$$

$$\tilde{W} = \begin{bmatrix} \tilde{U} \\ - \\ \tilde{V} \end{bmatrix}$$

$$\tilde{W} = \begin{bmatrix} \dot{i}_1^T \\ \vdots \\ \dot{i}_F^T \\ \dot{j}_1^T \\ \vdots \\ \dot{j}_F^T \end{bmatrix} \begin{bmatrix} s_1 & \dots & s_P \end{bmatrix} = RS$$

Motion
Shape
3XP
2FX3

Rank of S is 3, because points in 3D space are not Co-planar

Rank Theorem

Without noise, the registered measurement matrix $\tilde{W}$ is at most of rank three.

$$\tilde{W} = \begin{bmatrix} i_1^T \\ \vdots \\ i_F^T \\ j_1^T \\ \vdots \\ j_F^T \end{bmatrix} \begin{bmatrix} s_1 & \dots & s_p \end{bmatrix} = RS$$

How to find Translation?

$$\tilde{u}_{fp} = u_{fP} - a_f \quad \text{From (A)}$$

$$u_{fp} = \tilde{u}_{fP} + a_f \quad \tilde{u}_{fp} = i_f^T S_P \quad \text{From (B)}$$

$$u_{fp} = i_f S_p + a_f \quad u_{fp} = i_f^T (s_p - t_f)$$

$$a_f = -t_f^T i_f \quad \text{(D)} \quad \text{From (C)}$$

a_f is projection of camera translation along x-axis

Registered Measurement Matrix in terms of Motion and Shape

$$\tilde{u}_{fP} = i_f^T S_P \quad (\text{B})$$

$$\tilde{v}_{fP} = j_f^T S_P$$

$$\tilde{W} = \begin{bmatrix} \tilde{U} \\ - \\ \tilde{V} \end{bmatrix}$$

$$\tilde{W} = \begin{bmatrix} i_1^T \\ \vdots \\ i_F^T \\ j_1^T \\ \vdots \\ j_F^T \end{bmatrix} \begin{bmatrix} s_1 & \dots & s_P \end{bmatrix} = RS$$

3XP

2FX3

Rank of S is 3, because points in 3D space are not
Co-planar

Measurement Matrix in terms of Motion and Shape

$$u_{fp} = i_f^T (s_p - t_f) \quad \text{From (C)} \quad a_f = -t_f i_f^T \quad \text{(D)}$$

$$u_{fp} = i_f s_p + a_f \quad v_{fp} = j_f s_p + b_f$$

$$\mathbf{W} = \mathbf{R}\mathbf{S} + \mathbf{t}\mathbf{e}_p^T$$

$\begin{matrix} 2\text{FXP} & 2\text{FX3} & 3\text{XP} & 2\text{FX1} & 1\text{XP} \end{matrix}$

$$\mathbf{t} = (a_1, \dots, a_f, b_1, \dots, b_f)^T$$

$$\mathbf{e}_p^T = (1, \dots, 1)$$

Measurement Matrix in terms of Motion and Shape

$$u_{fp} = i_f s_p + a_f \quad v_{fp} = j_f s_p + b_f$$

$$\mathbf{W} = \mathbf{R}\mathbf{S} + \mathbf{t}\mathbf{e}_p^T$$

$\begin{matrix} 2FXP & 2FX3 & 3XP & 2FX1 & 1XP \end{matrix}$

$$W = \begin{bmatrix} u_{11} \dots u_{1p} \\ \vdots \\ u_{F1} \dots u_{FP} \\ v_{11} \dots v_{1p} \\ \vdots \\ v_{F1} \dots v_{FP} \end{bmatrix} = \begin{bmatrix} i_1 & j_1 & k_1 \\ \vdots & & \\ \vdots & & \\ i_F & j_F & k_F \end{bmatrix} \begin{bmatrix} X_1 & \dots & X_P \\ Y_1 & \dots & Y_P \\ Z_1 & \dots & Z_P \end{bmatrix} + \begin{bmatrix} a_1 \\ \vdots \\ a_F \\ b_1 \\ \vdots \\ b_F \end{bmatrix} [1 \quad \dots \quad 1]$$

Translation

Projected camera translation can be computed:

$$-i_f^T t_f = a_f = \frac{1}{P} \sum_{p=1}^P u_p$$

$$-j_f^T t_f = b_f = \frac{1}{P} \sum_{p=1}^P v_p$$

Noisy Measurements

- Without noise, the matrix $\tilde{W}$ must be at most of rank 3. When noise corrupts the images, however, $\tilde{W}$ will not be rank 3. Rank theorem can be extended to the case of noisy measurements.

Approximate Rank

SVD

$$\tilde{W} = O_1 \Sigma O_2$$

2FXP PXP PXP

Singular Value Decomposition (SVD)

- For some linear systems $Ax=b$, Gaussian Elimination or LU decomposition does not work, because matrix A is singular, or very close to singular. SVD will not only diagnose for you, but it will solve it.

Singular Value Decomposition (SVD)

Theorem: Any m by n matrix A , for which $m \geq n$, can be written as

$$A = O_1 \Sigma O_2$$

$m \times n \quad \quad m \times n \quad n \times n \quad n \times n$

Σ is diagonal

O_1, O_2 are orthogonal

$$O_1^T O_1 = O_2^T O_2 = I$$

Singular Value Decomposition (SVD)

If A is square, then O_1, Σ, O_2 are all square.

$$O_1^{-1} = O_1^T$$

$$O_2^{-1} = O_2^T$$

$$\Sigma^{-1} = \text{diag}\left(\frac{1}{w_j}\right)$$

$$\underset{n \times n}{A} = \underset{n \times n}{O_1} \underset{n \times n}{\Sigma} \underset{n \times n}{O_2}$$

Approximate Rank

$$\tilde{W} = O_1 \Sigma O_2$$

$$O_1 = \begin{bmatrix} \overset{3}{O_1'} & \overset{\mathbf{P}-3}{O_1''} \end{bmatrix} \quad \begin{matrix} 2\mathbf{F} \end{matrix}$$

$$\Sigma = \begin{bmatrix} \overset{3}{\Sigma'} & \overset{\mathbf{P}-3}{0} \\ 0 & \Sigma'' \end{bmatrix} \quad \begin{matrix} 3 \\ \mathbf{P}-3 \end{matrix}$$

$$O_1 \Sigma O_2 = O_1' \Sigma' O_2' + O_1'' \Sigma'' O_2''$$

$$O_2 = \begin{bmatrix} O_2' \\ O_2'' \end{bmatrix} \quad \begin{matrix} 3 \\ \mathbf{P}-3 \end{matrix}$$

Approximate Rank

$$\tilde{W} = O_1 \Sigma O_2 = O_1' \Sigma' O_2' + O_1'' \Sigma'' O_2''$$

$$\hat{W} = O_1' \Sigma' O_2'$$

The best rank 3 approximation to the ideal registered measurement matrix.

Rank Theorem for noisy measurement

The best possible shape and rotation estimate is obtained by considering only 3 greatest singular values of $\tilde{W}$ together with the corresponding left, right eigenvectors.

Approximate Rank

$$\hat{R} = O_1' [\Sigma']^{1/2}$$

Approximate Rotation matrix

$$\hat{S} = [\Sigma']^{1/2} O_2'$$

Approximate Shape matrix

$$\hat{W} = \hat{R} \hat{S}$$

This decomposition is not unique

$$\hat{W} = (\hat{R} Q)(Q^{-1} \hat{S})$$

Q is *any* 3X3 invertable matrix

Approximate Rank

$$R = \hat{R}Q$$

$$S = Q^{-1}\hat{S}$$

R and S are linear transformation of
approximate Rotation and shape matrices

How to determine Q ?

$$\hat{i}_f^T Q Q^T \hat{i}_f = 1$$

$$\hat{j}_f^T Q Q^T \hat{j}_f = 1$$

$$\hat{i}_f^T Q Q^T \hat{j}_f = 0$$

Rows of rotation matrix are unit vectors

Copyright Mubarak Shah 2003 and orthogonal

How to determine **Q**: Newton's Method

$$M\Delta\mathbf{q} = \varepsilon$$

$$f_1(\mathbf{q}) = \hat{i}_1^T Q Q^T \hat{i}_1^T - 1 = 0$$

$$f_2(\mathbf{q}) = \hat{j}_1^T Q Q^T \hat{j}_1^T - 1 = 0$$

$$f_3(\mathbf{q}) = \hat{i}_1^T Q Q^T \hat{j}_1^T = 0$$

$\vdots$

$$f_{3F-2}(\mathbf{q}) = \hat{i}_f^T Q Q^T \hat{i}_f^T - 1 = 0$$

$$f_{3F-1}(\mathbf{q}) = \hat{j}_f^T Q Q^T \hat{j}_f^T - 1 = 0$$

$$f_{3F}(\mathbf{q}) = \hat{i}_f^T Q Q^T \hat{j}_f^T = 0$$

$$\Delta\mathbf{q} = [\Delta q_1, \dots, \Delta q_9]$$

$$\mathbf{M}_{ij} = \frac{\partial f_i}{\partial q_j}$$

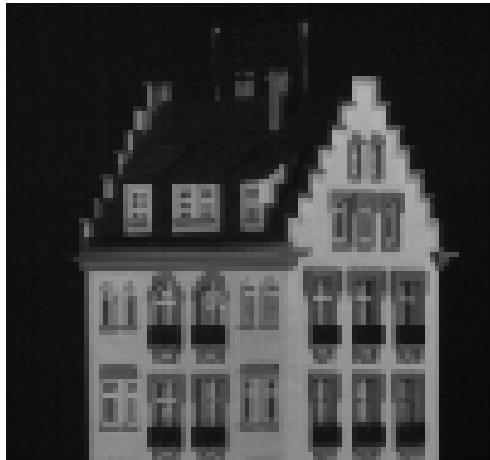
Jacobian

ε is error

Algorithm

- Compute SVD of $\tilde{W} = O_1 \Sigma O_2$
- define $\hat{R} = O_1' [\Sigma']^{1/2}$ $\hat{S} = [\Sigma']^{1/2} O_2'$
- Compute Q
- Compute $R = \hat{R} Q$ $S = Q^{-1} \hat{S}$

Hotel Sequence



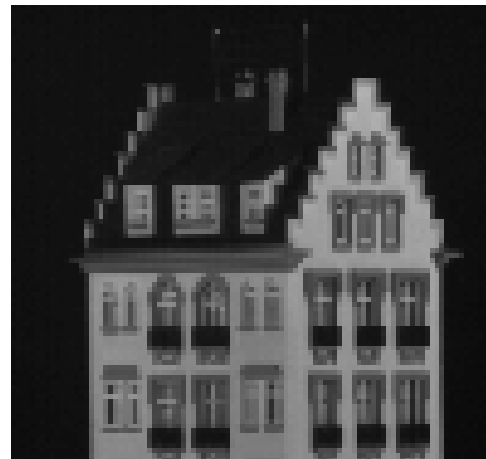
1



60

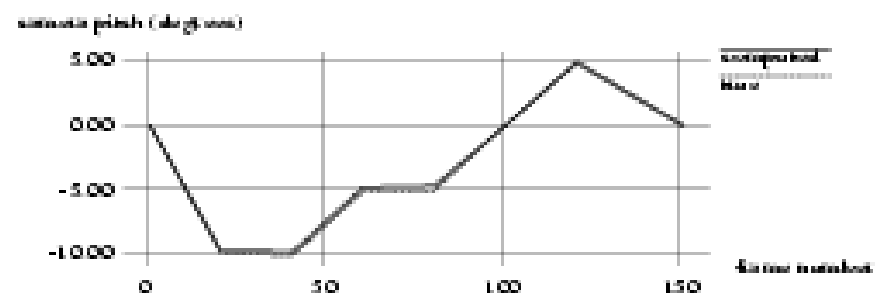
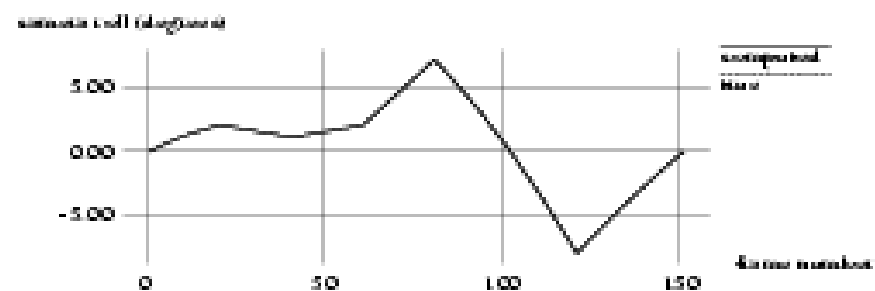
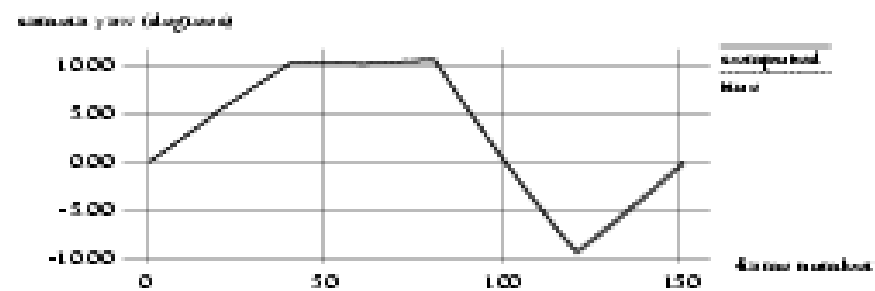


120

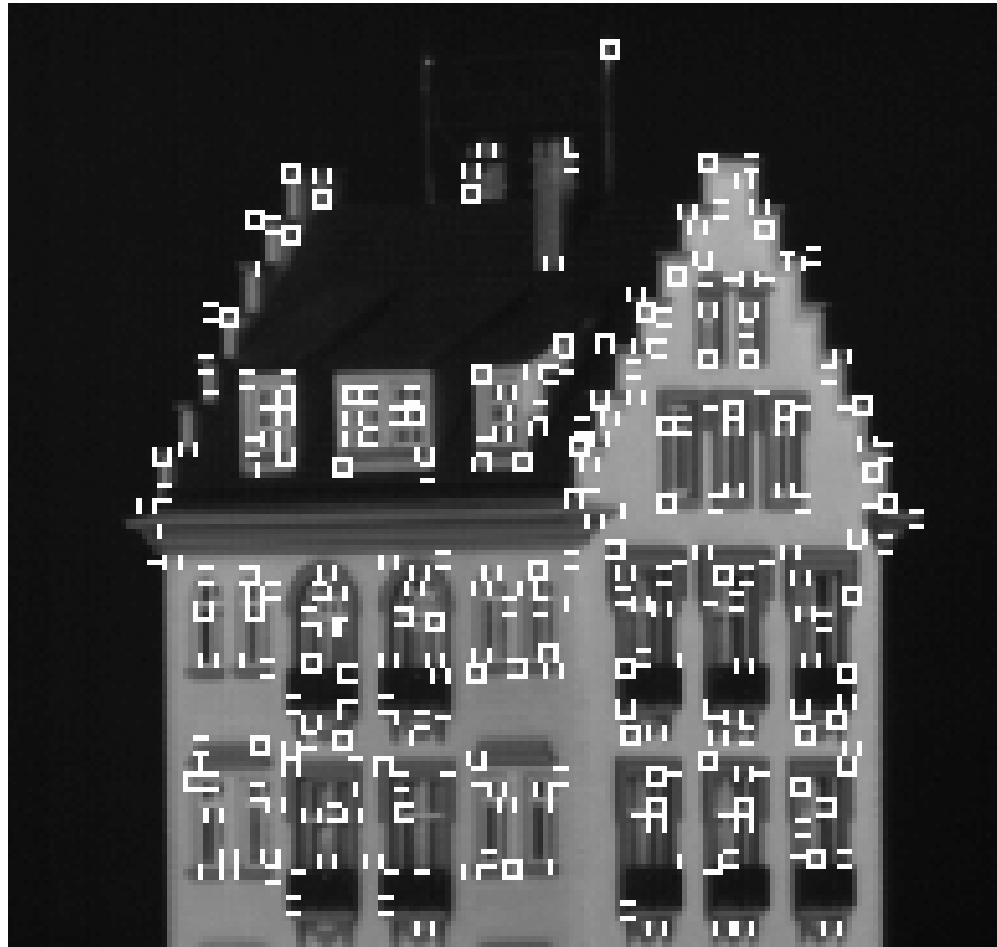


150

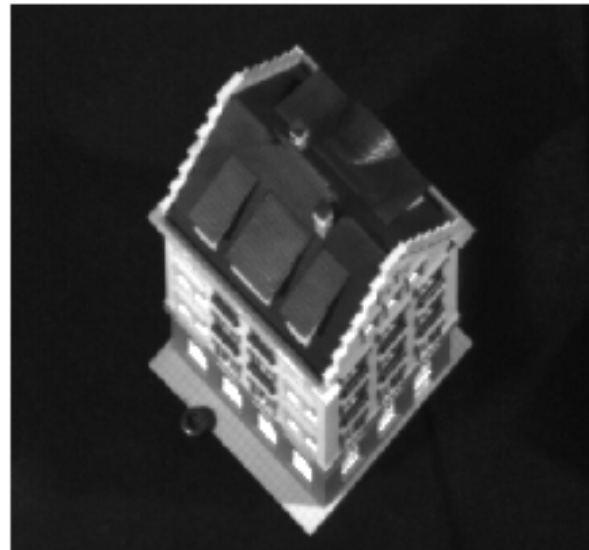
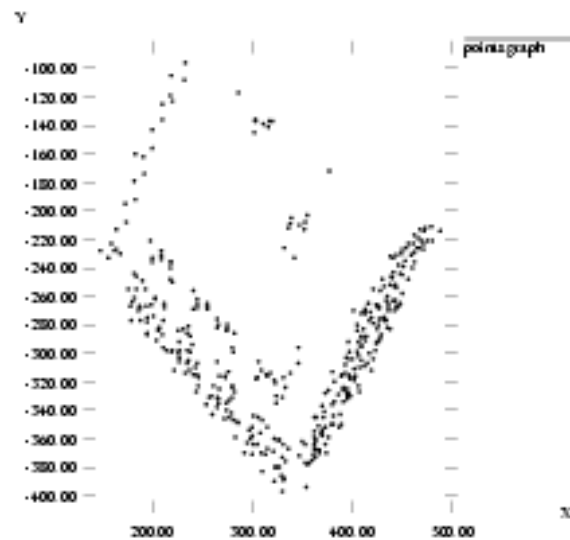
Results (rotations)



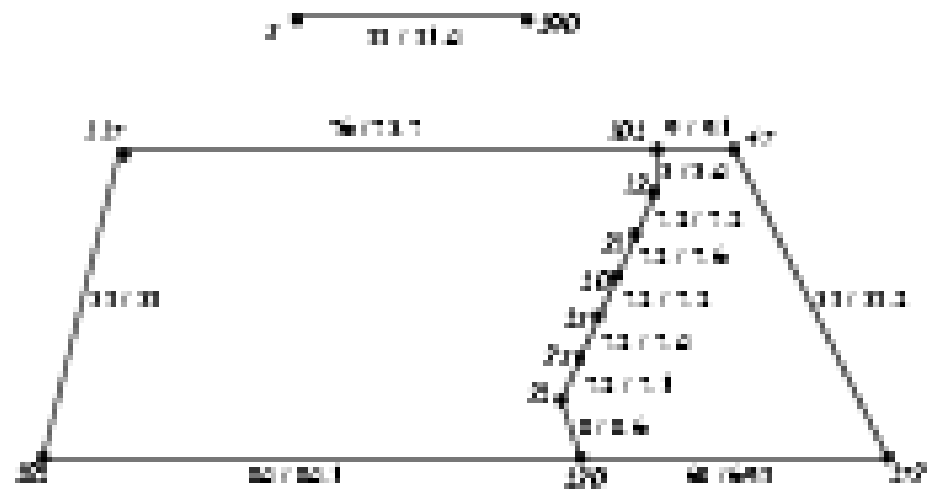
Selected Features



Reconstructed Shape



Comparison



House Sequence



1



60

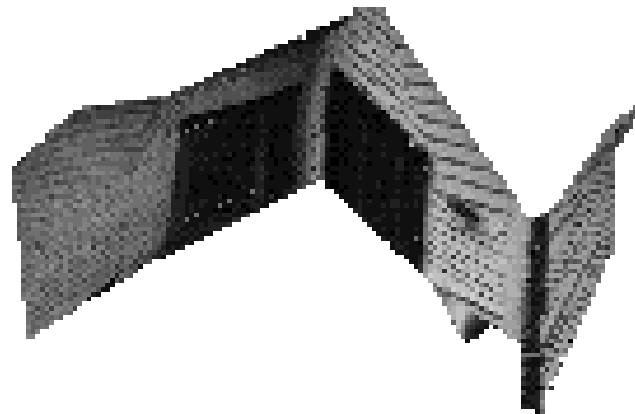


1000



1000

Reconstructed Walls



Paper

- C. Tomasi and T. Kanade. Shape and motion from image streams under orthography---a factorization method. *International Journal on Computer Vision*, 9(2):137-154, November 1992. ([2357 citations](#))